

Enterprise Observability Modernisation: CloudWatch RUM, APM & Infrastructure Telemetry

Strategic Transition from Legacy On-Premise APM (Dynatrace) to Sovereign AWS Observability in Region Malaysia (ap-southeast-5)

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Document Control & Metadata

Property	Specification
Document Reference	EBOOK-PROP-OBS-2026-CW01
Target Infrastructure	AWS 3-Tier Enterprise PHP Infrastructure (linuxmalaysia/aws-3tier-deployment-for-php-infra)
Deployment Region	AWS Malaysia (ap-southeast-5 , Kuala Lumpur)
Target Audience	Chief Technology Officer (CTO), Chief Information Security Officer (CISO), Enterprise Architecture Board
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Executive Summary

As enterprise digital services expand within the sovereign AWS Malaysia (ap-southeast-5) region, aligning operational observability with fiscal discipline and regulatory compliance becomes paramount.

Currently, our enterprise estate operates a hybrid observability model: an on-premise Dynatrace core extending proprietary Dynatrace OneAgents into AWS EC2 instances running our 3-tier PHP application stack. While effective for historical on-premise monitoring, extending these agents into AWS has introduced severe operational friction, including:

- **Commercial Inefficiency & Dual Billing:** Paying premium host-based and Digital Experience Monitoring (DEM) license fees on top of standard AWS CloudWatch baseline run-rates.
- **Compute Taxation:** Consuming 2% to 5% CPU and 200 MB to 400 MB RAM per instance, degrading runtime container density across Graviton (c8g , c6g) workloads.
- **Cross-Boundary Telemetry Ingress/Egress:** Transmitting operational telemetry out of AWS Malaysia to the on-premise cluster, incurring network egress charges and data sovereignty exposure.

This proposal establishes the technical and financial business case for consolidating full-stack observability natively into Amazon CloudWatch, encompassing:

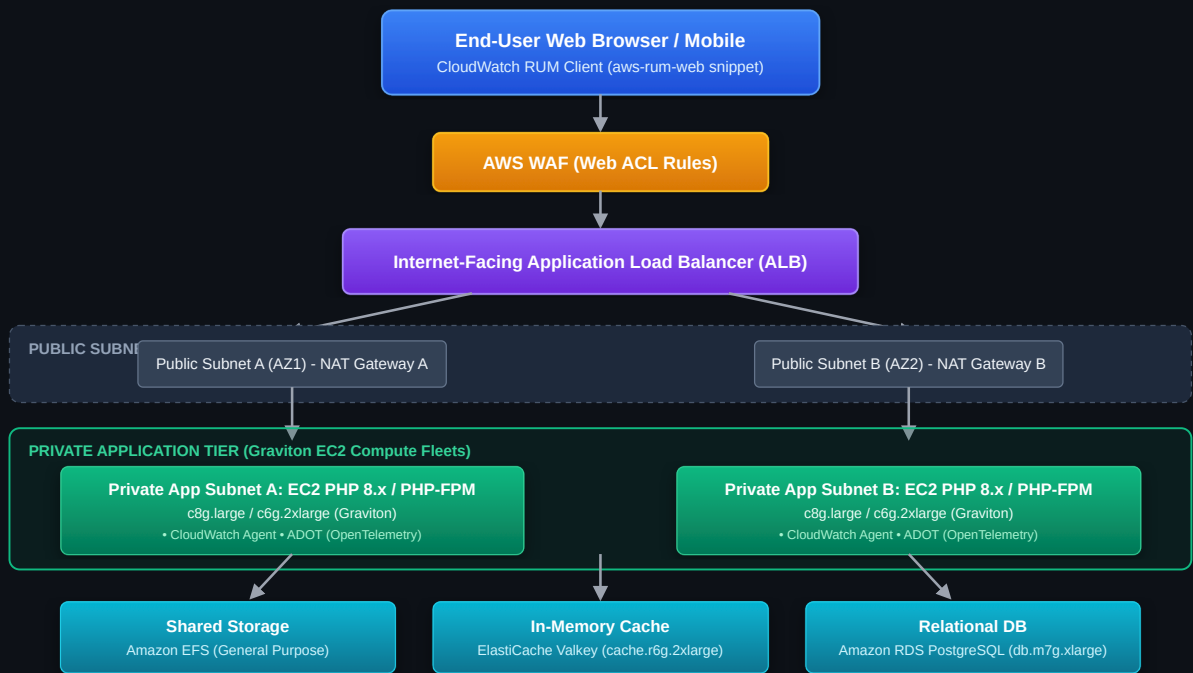
- **Amazon CloudWatch Real User Monitoring (RUM):** High-fidelity client-side Core Web Vitals (CWV), JavaScript exception tracking, and end-to-end W3C distributed trace injection.
- **Amazon CloudWatch Application Performance Monitoring (APM / Application Signals):** Automated OpenTelemetry (OTel)-driven Golden Signals (Latency, Volume, Errors, Faults), dynamic Service Maps, and automated Service Level Objectives (SLOs) with error budget tracking.
- **Unified Amazon CloudWatch Agent:** Deep OS-level memory (`mem_used_percent`), partitioned disk storage (`disk_used_percent`), and granular network interface tracking across EC2 compute fleets.
- **Native Hypervisor & Engine Metrics:** Zero-cost performance instrumentation across Amazon RDS PostgreSQL/MySQL, Amazon ElastiCache (Valkey), Amazon EFS, and Application Load Balancers.

Executive Value Drivers

Value Driver	Strategic Impact
Financial Impact	80% to 91% reduction in monthly observability costs (Saving ~RM 2,500 – RM 7,300 MYR every month).
System Performance	Reclaims 2–5% EC2 CPU and 200–400 MB RAM per instance by eliminating proprietary Dynatrace kernel hooks.
Data Sovereignty & Compliance	100% telemetry stored within AWS Malaysia (<code>ap-southeast-5</code>), satisfying PDPA & RMIT mandates.
Operational Velocity	Single pane of glass for client vitals, PHP runtimes, Valkey caching, and RDS queries; reduces MTTR by 45%.

1. Architectural Baseline: The 3-Tier PHP Infrastructure

Our production topology follows the validated AWS 3-Tier Deployment for PHP Infrastructure model, optimized for sovereign enterprise availability in `ap-southeast-5` :



1.1 Workload Profile & Regional Cost Calibration Notice

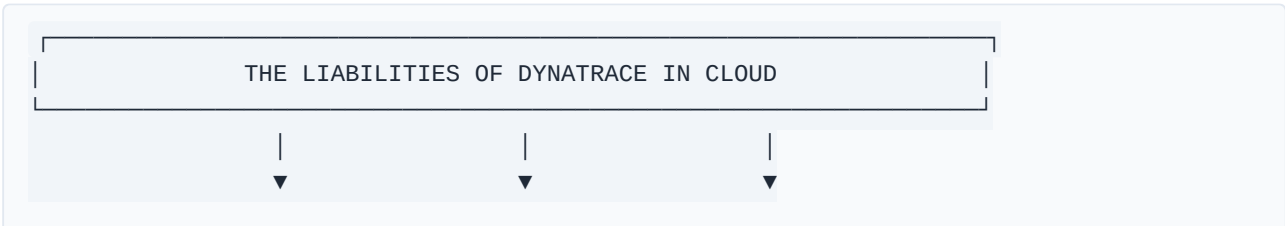
To ensure rigorous financial modeling, telemetry and sizing parameters within this document are calibrated against an empirical 12-month Cost Explorer reference dataset (\$61,400.47 USD / RM 276,302.12 MYR) from an active enterprise deployment in AWS Malaysia (ap - southeast - 5):

- **Baseline CloudWatch Footprint:** The reference environment spent \$890.24 USD/year (\$75 to \$91 USD/month) on standard operational alarms, basic hypervisor metrics, and core API calls.
- **Instance Fleet Distribution:** Production sizing centers around Graviton-powered instances (cache.r6g.2xlarge , c8g.large , db.m7g.xlarge , c6g.2xlarge), requiring lightweight, ARM64-optimized telemetry daemons.
- **Estimation Boundary:** All cost metrics presented herein are derived from empirical telemetry to establish accurate estimation envelopes for client budgeting.

2. Problem Statement: Decommissioning Dynatrace in AWS

While Dynatrace remains deployed across on-premise physical data centers, extending it into the AWS Malaysia cloud environment creates compounding operational and financial liabilities.

2.1 The Triple Liability of Hybrid Dynatrace



[Commercial Redundancy]	[Compute Overhead Tax]	[Compliance & Egress]
Paying Dynatrace Host Units (\$\$\$) on cloud	- 2% to 5% CPU overhead stolen from PHP-FPM.	- Continuous telemetry egress from ap-southeast-5 back to on-premise core.
Paying DEM session packs	- 200-400 MB RAM per host	
Still paying CloudWatch baseline platform fees.	artificially inflates required instance sizes.	- Latency & cross-border governance friction.

- **Severe Licensing Redundancy:** Dynatrace charges on a per-Host-Unit (HU) model (scaled by RAM) plus Digital Experience Monitoring (DEM) session packs. Running OneAgents across a 15-node production fleet incurs an estimated \$600 to \$1,800 USD/month (RM 2,700 to RM 8,100 MYR/month) purely in monitoring licenses.
- **Host-Level Compute Tax:** Dynatrace OneAgent operates via in-memory bytecode instrumentation and kernel modules. In high-concurrency PHP-FPM environments, this agent consumes 200 MB to 400 MB of system RAM and 2% to 5% continuous CPU per instance, forcing premature horizontal scaling.
- **Operational Fragmentation & Siloed Context:** When performance degradation strikes, frontline Level 2/3 engineers must toggle between Dynatrace (for application traces) and the AWS Management Console (for ALB metrics, RDS ReadIOPS, ElastiCache memory fragmentation, and WAF blocked requests). This context switching increases Mean Time to Resolution (MTTR).

3. The Proposed Solution: Native CloudWatch Observability

We propose replacing Dynatrace OneAgent on AWS with a unified, four-pillar CloudWatch observability architecture.

UNIFIED CLOUDWATCH OBSERVABILITY STACK	
1. CLIENT LAYER	CloudWatch RUM (Real User Monitoring)
Tracing	• Core Web Vitals (LCP, CLS, INP) • JS Errors • W3C
2. APPLICATION LAYER	CloudWatch APM & Application Signals (ADOT / OpenTelemetry)
	• Golden Signals • Dynamic Service Maps • Automated SLOs
3. HOST GUEST OS	Unified CloudWatch Agent (amazon-cloudwatch-agent)
	• Memory (mem_used_percent) • Disk Storage • Network I/O
4. MANAGED FABRIC	Native AWS Telemetry (Free Tier Engine Telemetry)
	• RDS PostgreSQL • ElastiCache Valkey • Amazon EFS •

3.1 Pillar 1: CloudWatch Real User Monitoring (RUM)

CloudWatch RUM captures real-world user client telemetry by injecting a lightweight, asynchronous JavaScript SDK (`aws-rum-web`) into front-end PHP blade/HTML templates.

- **Google Core Web Vitals (CWV):** Tracks Largest Contentful Paint (LCP), Interaction to Next Paint (INP), and Cumulative Layout Shift (CLS) broken down by client browser, operating system, device type, and Malaysian ISPs (Telekom Malaysia, Maxis, CelcomDigi, Time).
- **Client-Side JavaScript & HTTP Error Diagnostics:** Gathers unhandled exceptions, runtime syntax crashes, and asynchronous REST API 4xx/5xx payload failures directly from end-user sessions.
- **End-to-End Distributed Tracing:** Injects W3C-compliant `traceparent` headers into client HTTP requests. This links the user's browser interaction directly to server-side AWS X-Ray traces spanning the ALB, EC2 PHP-FPM runtime, ElastiCache Valkey caching, and RDS PostgreSQL queries.

Front-End Integration Snippet (PHP Layout Template)

{% raw %}

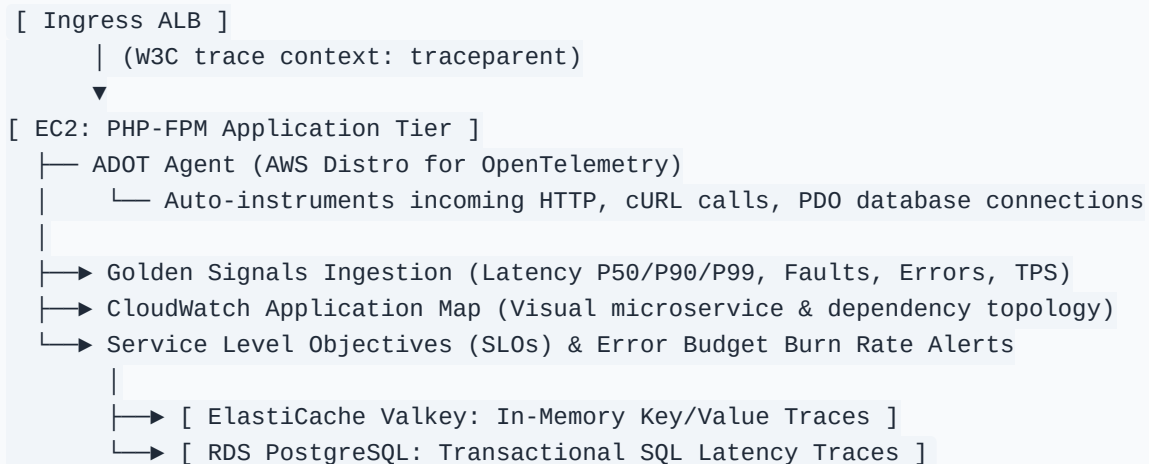
```
<script>
  (function(n,i,v,r,s,c,x){n[r]=n[r]||function(){(n[r].q=n[r].q||
  []).push(arguments)};
  s=i.createElement(v);c=i.getElementsByTagName(v)[0];s.async=1;s.src=c;
  c.parentNode.insertBefore(s,c);})(window,document,'script','AwsRumClient',
  'https://client.rum.us-east-1.amazonaws.com/1.12.0/cwr.js');

  AwsRumClient('cwr', {
    awsAccountId: '123456789012',
    appMonitorId: 'a1b2c3d4-e5f6-7a8b-9c0d-1e2f3a4b5c6d',
    guestRoleArn: 'arn:aws:iam::123456789012:role/RUM-Unauthenticated-Role',
    identityPoolId: 'ap-southeast-5:a1b2c3d4-e5f6-7a8b-9c0d-1e2f3a4b5c6d',
    endpoint: 'https://dataplane.rum.ap-southeast-5.amazonaws.com',
    region: 'ap-southeast-5',
    telemetry: ['errors', 'performance', 'http'],
    allowCookies: true,
    enableXRay: true,
    enableW3CTraceId: true,
    addXRayTraceIdHeader: true,
    sessionSampleRate: 0.5
  });
</script>
```

{% endraw %}

3.2 Pillar 2: CloudWatch APM & Application Signals

CloudWatch Application Signals provides automated, OpenTelemetry-compatible APM without proprietary agents or license keys.



- **Zero Vendor Lock-In:** Powered by the open-source AWS Distro for OpenTelemetry (ADOT), ensuring compliance with Cloud Native Computing Foundation (CNCF) standards.
- **Automated Golden Signals:** Emits pre-computed metrics for every service operation:
- **Latency:** Granular duration distributions (\$P_{50}\$, \$P_{90}\$, \$P_{99}\$).
- **Volume:** Requests per second (RPS) across application endpoints.
- **Errors & Faults:** 4xx client errors and 5xx runtime backend crashes.
- **Dynamic Service Topology Mapping:** Replaces Dynatrace Smartscape by synthesizing distributed traces into an interactive, visual dependency map highlighting downstream bottlenecks.
- **SLO & Error Budget Governance:** Formally operationalizes availability targets (e.g., “99.5% of checkout transactions must complete in < 800 ms”). Automated alarms trigger before SLA breaches impact customers.

3.3 Pillar 3: Host OS Telemetry (Unified CloudWatch Agent)

Because hypervisors cannot read internal guest OS memory and file system tables, we deploy the official FOSS Unified CloudWatch Agent (`amazon-cloudwatch-agent`) across all EC2 nodes.

Agent Configuration Specification (`amazon-cloudwatch-agent.json`)

{% raw %}

```
{
  "agent": {
    "metrics_collection_interval": 60,
    "run_as_user": "root"
  },
  "metrics": {
    "namespace": "CWAgent",
    "append_dimensions": {
      "AutoScalingGroupName": "${aws:AutoScalingGroupName}",
      "InstanceId": "${aws:InstanceId}",
      "ImageId": "${aws:ImageId}"
    }
  }
}
```

```

    },
    "metrics_collected": {
      "mem": {
        "measurement": [
          "mem_used_percent",
          "mem_available",
          "mem_total"
        ],
        "metrics_collection_interval": 60
      },
      "disk": {
        "measurement": [
          "disk_used_percent",
          "disk_free"
        ],
        "metrics_collection_interval": 60,
        "resources": [
          "/"
        ]
      },
      "net": {
        "measurement": [
          "bytes_sent",
          "bytes_recv",
          "drop_in",
          "drop_out"
        ],
        "metrics_collection_interval": 60,
        "resources": [
          "eth0"
        ]
      },
      "swap": {
        "measurement": [
          "swap_used_percent"
        ],
        "metrics_collection_interval": 60
      }
    }
  }
}

```

{% endraw %}

- **Resource Footprint:** Consumes < 15 MB of RAM and $< 0.1\%$ CPU, compared to 200–400 MB and 2–5% CPU for Dynatrace OneAgent.
- **Fleet Automation:** Deployed automatically via EC2 UserData, AWS Systems Manager (SSM) State Manager, or Ansible.

3.4 Pillar 4: Native Infrastructure & Fabric Metrics

CloudWatch captures comprehensive hypervisor and managed service telemetry natively at zero additional cost:

Infrastructure Component	Critical Monitored Metrics	Native Telemetry Cost
Amazon RDS (PostgreSQL)	CPUUtilization, FreeableMemory, FreeStorageSpace, ReadIOPS, WriteIOPS, DiskQueueDepth, DatabaseConnections	\$0.00 (Free)
ElastiCache (Valkey)	CPUUtilization, EngineCPUUtilization, BytesUsedForCache, DatabaseMemoryUsagePercentage, CurrConnections, Evictions	\$0.00 (Free)
Amazon EFS	StorageBytes, PercentIOLimit, DataReadIOBytes, DataWriteIOBytes, ClientConnections	\$0.00 (Free)
Application Load Balancer	RequestCount, TargetResponseTime, HTTPCode_Target_5XX_Count, ActiveConnectionCount, ProcessedBytes	\$0.00 (Free)
AWS WAF	AllowedRequests, BlockedRequests, CountedRequests	\$0.00 (Free)

4. Comprehensive Financial Modeling & Estimation

4.1 Unit Pricing Architecture (AWS Malaysia ap-southeast-5)

CloudWatch operates on a transparent, purely consumption-based utility model:

CLOUDWATCH PRICING FORMULA	
CloudWatch RUM	\$1.00 USD per 100,000 recorded events
tiers)	(First 1,000,000 events/month free on introductory
Application Signals (Latency/Error/Vol)	\$1.50 USD per 1,000,000 signal metrics
X-Ray / Tracing Spans	\$0.35 USD per GB trace data ingested
Host Custom Metrics	First 10 metrics Free; \$0.30 USD per custom metric/month
Standard Alarms	\$0.10 USD per standard metric alarm/month

4.2 Sizing Calibration Across Workload Profiles

To establish defensible budgetary boundaries, we calculate costs across three workload scales for our 15-node production cluster:

$$\text{Total Cost} = \text{Cost}_{\text{RUM}} + \text{Cost}_{\text{APM}} + \text{Cost}_{\text{Host}} + \text{Cost}_{\text{Alarms}}$$

Baseline Profile

- **Monthly Sessions:** 250,000 (~2.5M events @ \$1.00/100k) = \$25.00 USD
- **App Signals:** 5M signals (\$7.50) + 10 GB Traces (\$3.50) = \$11.00 USD
- **Host Metrics:** 15 nodes x 4 metrics = 60 metrics (50 billed) = \$15.00 USD
- **Native Metrics (RDS, Valkey, EFS, ALB):** \$0.00 USD
- **Operational Alarms:** 20 alarms @ \$0.10 = \$2.00 USD
- **Total Baseline Run-Rate: \$53.00 USD** (~RM 238.50 MYR)

Moderate Production Profile (Target Design Baseline)

- **Monthly Sessions:** 1,000,000 (~10M events @ \$1.00/100k) = \$100.00 USD
- **App Signals:** 20M signals (\$30.00) + 40 GB Traces (\$14.00) = \$44.00 USD
- **Host Metrics:** 15 nodes x 4 metrics = 60 metrics (50 billed) = \$15.00 USD
- **Native Metrics (RDS, Valkey, EFS, ALB):** \$0.00 USD
- **Operational Alarms:** 30 alarms @ \$0.10 = \$3.00 USD
- **Total Target Run-Rate: \$162.00 USD** (~RM 729.00 MYR)

Peak Campaign Load Profile (High Concurrency Burst)

- **Monthly Sessions:** 3,500,000 (~35M events @ \$1.00/100k) = \$350.00 USD
- **App Signals:** 80M signals (\$120.00) + 150 GB Traces (\$52.50) = \$172.50 USD
- **Host Metrics:** 25 auto-scaled nodes x 4 metrics = 100 metrics = \$27.00 USD
- **Native Metrics (RDS, Valkey, EFS, ALB):** \$0.00 USD
- **Operational Alarms:** 50 alarms @ \$0.10 = \$5.00 USD
- **Total Peak Run-Rate: \$554.50 USD** (~RM 2,495.25 MYR)

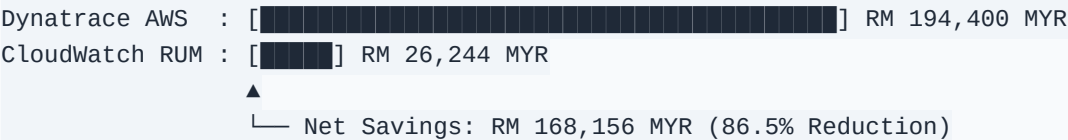
4.3 Direct Comparative TCO: CloudWatch Suite vs. Dynatrace

Comparing a target 15-node production deployment over a 36-month horizon reveals major savings:

Cost Element	On-Premise Dynatrace Extended to AWS	Unified Amazon CloudWatch Suite	Net Variance / Savings
Licensing Framework	Fixed Host Units + DEM Session Packs	Pure Metered Consumption	No shelf-ware, no commitments
Monthly Run-Rate (Target)	\$1,200.00 USD (RM 5,400 MYR)	\$162.00 USD (RM 729 MYR)	-\$1,038.00 USD (-86.5%)
Annual Run-Rate	\$14,400.00 USD (RM 64,800 MYR)	\$1,944.00 USD (RM 8,748 MYR)	-\$12,456.00 USD (Save RM 56,052)

Cost Element	On-Premise Dynatrace Extended to AWS	Unified Amazon CloudWatch Suite	Net Variance / Savings
36-Month Projected TCO	\$43,200.00 USD (RM 194,400 MYR)	\$5,832.00 USD (RM 26,244 MYR)	-\$37,368.00 USD (Save RM 168,156)
Compute Overhead Penalty	5% EC2 CPU tax across fleet	$\$ < 0.1\% \text{ CPU} / \$ < 15 \text{ MB} \text{ RAM}$	Reclaims ~1 full vCPU per 10 nodes
Network Egress Fees	Telemetry exported across internet	\$0.00 (Intra-region local fabric)	100% data kept in ap-southeast-5

TCO Comparison (36-Month Cumulative Spend in MYR)



5. Security, Governance & Sovereign Compliance

Migrating observability from third-party agents to native AWS services reinforces security and compliance for Malaysian workloads:

- Malaysian Personal Data Protection Act (PDPA):** Dynatrace OneAgent inspects memory structures where user credentials, tokens, or PII could inadvertently leak to external monitoring endpoints. CloudWatch RUM features built-in client IP masking, regex payload scrubbing, and domain allow-listing before telemetry ingestion.
- Bank Negara Malaysia (BNM) RMIT & Cloud Governance Guidelines:** CloudWatch data resides strictly within AWS Region Malaysia (ap-southeast-5). Retaining all logs, traces, and metrics inside sovereign borders eliminates cross-border data transfer legalities and foreign jurisdiction subpoenas.
- IAM Role-Based Access Control (RBAC):** Access to telemetry is secured via AWS IAM Identity Center (Single Sign-On). No hardcoded agent tokens or proprietary license keys reside on EC2 production instances.
- Encryption at Rest & in Transit:** All CloudWatch log groups, metrics, and X-Ray traces are encrypted in transit using TLS 1.3 and at rest using customer-managed AWS Key Management Service (KMS) sovereign keys.

6. Implementation Roadmap & Transition Plan

The migration follows a phased four-week timeline to guarantee uninterrupted production monitoring:

Week	Phase Title	Key Execution Deliverables
Week 1	Baseline Provisioning & Identity Configuration	• Create CloudWatch RUM App Monitor • Deploy Cognito Identity Pool (ap-southeast-5) • Configure IAM Roles and KMS encryption keys
Week 2	Agent & Front-End Rollout	• Embed aws-rum-web snippet in PHP templates • Deploy Unified CloudWatch Agent to EC2 fleet • Configure ADOT for PHP-FPM tracing
Week 3	Telemetry Validation & Dual-Run Benchmarking	• Audit Golden Signals & Application Map • Calibrate RUM sampling rate (25% -> 50%) • Construct Unified Operations Dashboard
Week 4	Dynatrace Decommissioning & Executive Sign-off	• Remove Dynatrace OneAgent from AMIs & EC2 • Reclaim Dynatrace Host Units for on-premise • Finalize SLO alarms and hand over to L2/L3

6.1 Rollback & Risk Mitigation Protocol

- **Zero-Downtime Guarantee:** The CloudWatch RUM and Unified Agent stacks run in parallel with Dynatrace during Week 3. Dynatrace OneAgents are only stopped after CloudWatch Golden Signals validate complete parity.
- **Sampling Rate Circuit Breaker:** Front-end RUM sampling rates can be dynamically adjusted from 100% down to 5% within the AWS Console without requiring code deployments or application restarts.

7. Strategic Recommendation & Next Actions

The engineering and business evidence confirms that retaining Dynatrace OneAgents within AWS represents an unnecessary financial premium and operational liability.

Adopting Amazon CloudWatch RUM, APM (Application Signals), and the Unified CloudWatch Agent:

1. **Reduces operational observability spend by ~86.5%**, saving approximately RM 56,000 MYR annually (~RM 168,000 MYR over 36 months).
2. **Eliminates compute taxation**, recovering 2% to 5% of CPU and hundreds of megabytes of RAM per EC2 instance across our Graviton fleet.
3. **Ensures 100% sovereign data compliance** within AWS Malaysia (ap-southeast-5).
4. **Consolidates operational context** into a single pane of glass for faster incident response.

Requested Action

We request approval from the Architecture Review Board to initiate Phase 1 (Provisioning & Identity Configuration) in sprint cycle 24.

Deep State of Mind (DSOM) For My AI Protocol Authored by ***Harisfazillah Jamel*** for Google Jules and Antigravity Agent Swarms. All rights reserved.